# EN3160 – Fitting and Alignment

## Assignment 2 Report

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## 1. Laplacian-of-Gaussian blob detection

The supplied sunflower photograph is loaded with `cv2.IMREAD_REDUCED_COLOR_4`, giving a $360 \times 360$ image. For each Gaussian scale, the detector evaluates the absolute scale-normalised Laplacian,

$$L(x, y, \sigma) = \left|\sigma^2 \nabla^2 (G_\sigma * I)(x, y)\right|, \qquad r \approx \sqrt{2}\,\sigma.$$

Local maxima are extracted in the spatial-scale response stack and duplicate scale-space candidates are suppressed. To focus on sunflower heads rather than leaves and tree texture, candidates are restricted to $y \geq 180$ and validated by the yellow-petal fraction and grayscale contrast of an annulus around a darker centre. No `cv2.HoughCircles` call is used.

**Selected parameters.** 32 logarithmically spaced scales, $\sigma = 0.5 - -8.0$ reduced-image pixels; response threshold 0.06; spatial neighbourhood $7 \times 7$; yellow annulus fraction $\geq 0.28$; ring-minus-centre grayscale contrast $\geq 8.0$. The output draws every validated detection, not just the largest few.

**Figure 1:** LoG detections over the course sunflower image. All 271 validated candidates are drawn.

**Largest detected circles**

Centres and radii use reduced-image pixels; multiply by four to convert to original-image coordinates.

| # | $x$ | $y$ | $\sigma$ | radius | resp. |
|---|---|---|---|---|---|
| 1 | 282 | 338 | 7.32 | 10.35 | 0.349 |
| 2 | 79 | 258 | 6.69 | 9.46 | 0.178 |
| 3 | 106 | 255 | 6.12 | 8.65 | 0.362 |
| 4 | 179 | 327 | 6.12 | 8.65 | 0.345 |
| 5 | 178 | 261 | 5.59 | 7.91 | 0.353 |
| 6 | 339 | 206 | 5.59 | 7.91 | 0.098 |
| 7 | 29 | 272 | 5.12 | 7.23 | 0.151 |
| 8 | 189 | 240 | 4.68 | 6.62 | 0.362 |

**Interpretation.** Extending the scale range down to 0.5 px recovers smaller flower heads that were missed by the original 1.3 px lower limit. Colour/contrast validation rejects many foliage responses. The detector still finds candidate blobs rather than semantically segmenting every flower: occlusion, overlapping heads, shadows and similar leaf texture can cause misses or false positives.

```
response = abs(sigma**2 * gaussian_laplace(gray, sigma))
peaks = local maxima in the spatial-scale response stack
radius = sqrt(2) * sigma
```

## 2. RANSAC line and circle fitting

Listing 1 is reproduced with a fixed random seed and 100 points: 50 noisy circle points with ground truth centre $(2, 3)$ and radius 10 (radial noise standard deviation 0.625), and 50 noisy line points from $y = -x+2$ (vertical noise standard deviation 1). The hand-coded RANSAC line model uses a unit normal $[a, b]^T$ and perpendicular distance residuals. Its consensus is refitted using total least squares; those inliers are removed before circle RANSAC uses three-point samples and minimises radial residuals on the consensus.

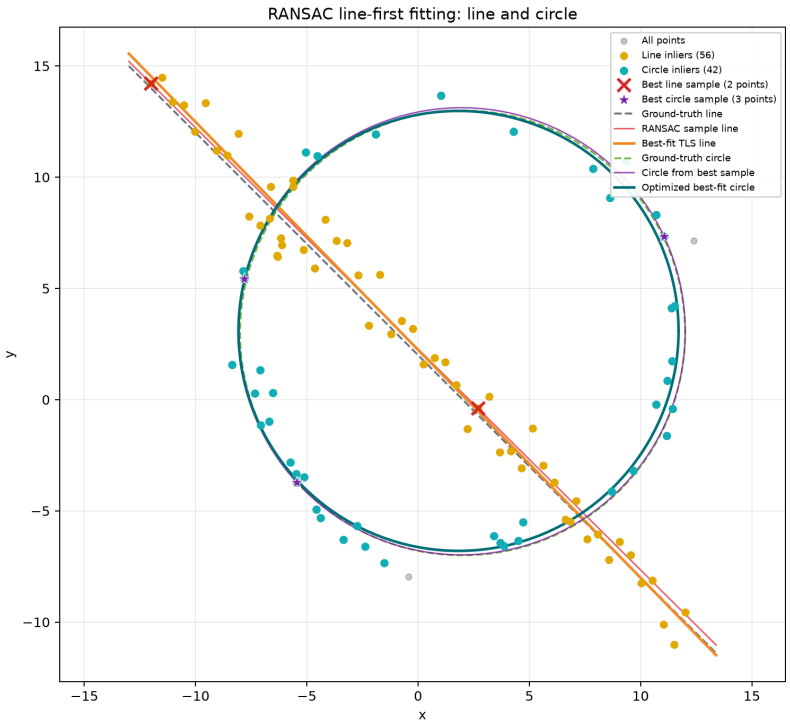



**Figure 2:** Point set, ground-truth and fitted models, selected minimal samples and consensus sets.

**RANSAC settings**

| | |
|---|---|
| Line | 2,500 iterations; threshold 1.7; minimum consensus 35 |
| Circle | 4,500 iterations; threshold 1.25; minimum consensus 32 |

**Estimated models**

| | |
|---|---|
| Line normal | $(0.715649, 0.698460)$ |
| Normal norm | 1.000000 |
| Line $d$ | 1.561775 |
| Line $y = mx + b$ | $m = -1.024609$, $b = 2.236025$ |
| Circle centre | $(1.817367, 3.089484)$ |
| Radius | 9.888575 |

### Ground truth versus estimate

| Parameter | Ground truth | Estimate |
|---|---|---|
| Line slope | $-1.0000$ | $-1.024609$ |
| Line intercept | 2.0000 | 2.236025 |
| Circle centre | $(2, 3)$ | $(1.817367, 3.089484)$ |
| Circle radius | 10.0000 | 9.888575 |

**Consensus and interpretation.** The line-first fit returns 56/100 line inliers (50 true-line + 6 true-circle points). After removing them, the circle fit returns 42 circle inliers (42 true-circle + 0 true-line points). The estimated circle centre is approximately 0.204 units from the true centre and the radius error is approximately 0.111. The fitted line is close to the ground truth; residual differences arise from noise and random sampling.

**(d) If the circle is fitted first.** Circle-first RANSAC returns 57 points (48 true-circle and 9 line points). The remaining 43 points then yield 41 true-line inliers. Therefore, model order changes the assignment of ambiguous points: the circle can absorb points from the line before the line fit ever sees them. Sequential RANSAC is effective here because the selected line consensus is removed before estimating the circle.

## 3. Homography and perspective overlay

A homography projects a planar overlay into a four-corner quadrilateral in the background. For each example, source corners are the overlay image corners; destination points are clicked or specified in the clockwise order top-left, top-right, bottom-right, bottom-left. The projective transform is computed with `cv2.getPerspectiveTransform` and applied with `cv2.warpPerspective`. A warped validity mask limits blending to the overlay area so unfilled pixels do not blacken the background.

**(a)** Union Jack on the Wadham College facade. **(b)** Sunflower image placed as a small poster.

**Figure 3:** Two overlays warped onto planar regions in the same supplied architecture image.

### Destination quadrilaterals (TL, TR, BR, BL)

| Setting | Example 1: flag | Example 2: poster |
|---|---|---|
| Background | `wadham.jpg` | `wadham.jpg` |
| Overlay | `flag.png` | sunflower field photo |
| Coordinates | $(220, 250), (480, 277), (474, 410), (215, 394)$ | $(72, 295), (185, 303), (181, 382), (69, 374)$ |
| Blend $\alpha$ | 0.97 | 0.97 |

**Rationale and discussion.** A flag is a recognisable graphic for a large facade area, whereas the sunflower photograph demonstrates a smaller poster. The projected corners follow each selected quadrilateral, and the validity mask preserves the rest of the architecture. Correct point order is crucial; misordered corners can rotate, flip or twist the overlay. A single homography models one plane, so it cannot accurately handle protruding objects or a target surface with substantial depth variation. The interactive script also permits selecting four new corners for other image pairs.

## 4. SIFT PCB alignment and difference images

The reference is `AECC1.jpg`. SIFT keypoints/descriptors are matched by `cv2.BFMatcher` using the L2 metric. Lowe's ratio test (0.75) removes ambiguous pairs; RANSAC estimates a homography with a 5 px reprojection threshold. Each test PCB is warped to the template dimensions. The absolute grayscale difference is computed only in the valid warped overlap, excluding uncovered black borders from its statistics.

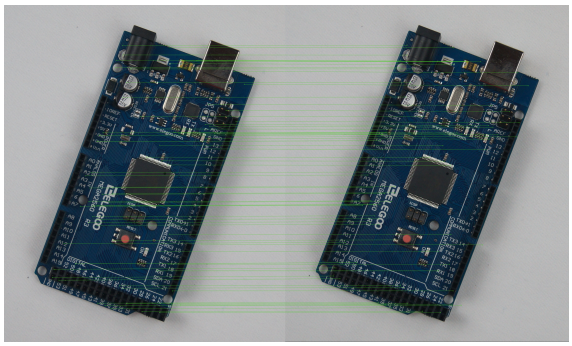
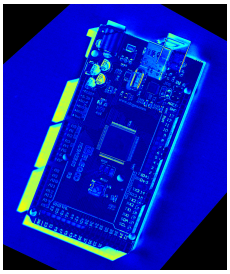


**Figure 4:** Example SIFT inlier matches: AECC2 to AECC1.

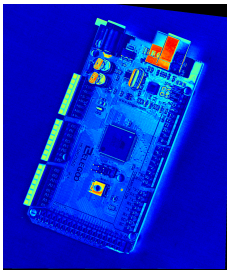
**Alignment and grayscale difference metrics**

| Image | Matches | Inliers | Mean $\lvert\Delta\rvert$ | 95th pct. |
|---|---|---|---|---|
| AAAA1 | 51 | 18 | 28.8 | 107 |
| ADAD1 | 369 | 194 | 31.2 | 92 |
| AECC2 | 4322 | 3847 | 3.5 | 9 |
| AECC3 | 5629 | 5006 | 3.2 | 8 |
| AECC4 | 3830 | 3402 | 4.0 | 10 |

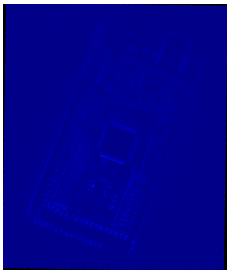
**Difference heatmaps** (all test images; brighter/hotter regions indicate larger local differences)



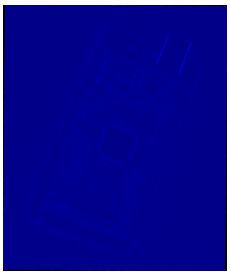
AAAA1



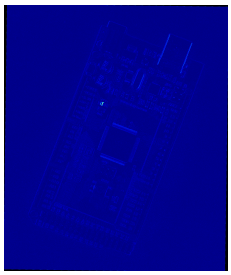
ADAD1



AECC2



AECC3



AECC4

**Interpretation.** AAAA1 and ADAD1 have substantially larger mean differences than AECC2–AECC4, whose mean differences are approximately 3–4 grayscale levels. A high difference around a USB Type-B port, capacitors or the reset button can be caused by changed/missing components, small position or shape changes, illumination, resampling after warping, or residual registration error. A homography corrects global planar perspective but cannot remove genuine board/component differences or local parallax. Thus the difference map is a comparison aid, not by itself proof of a manufacturing fault.